INCH POUND

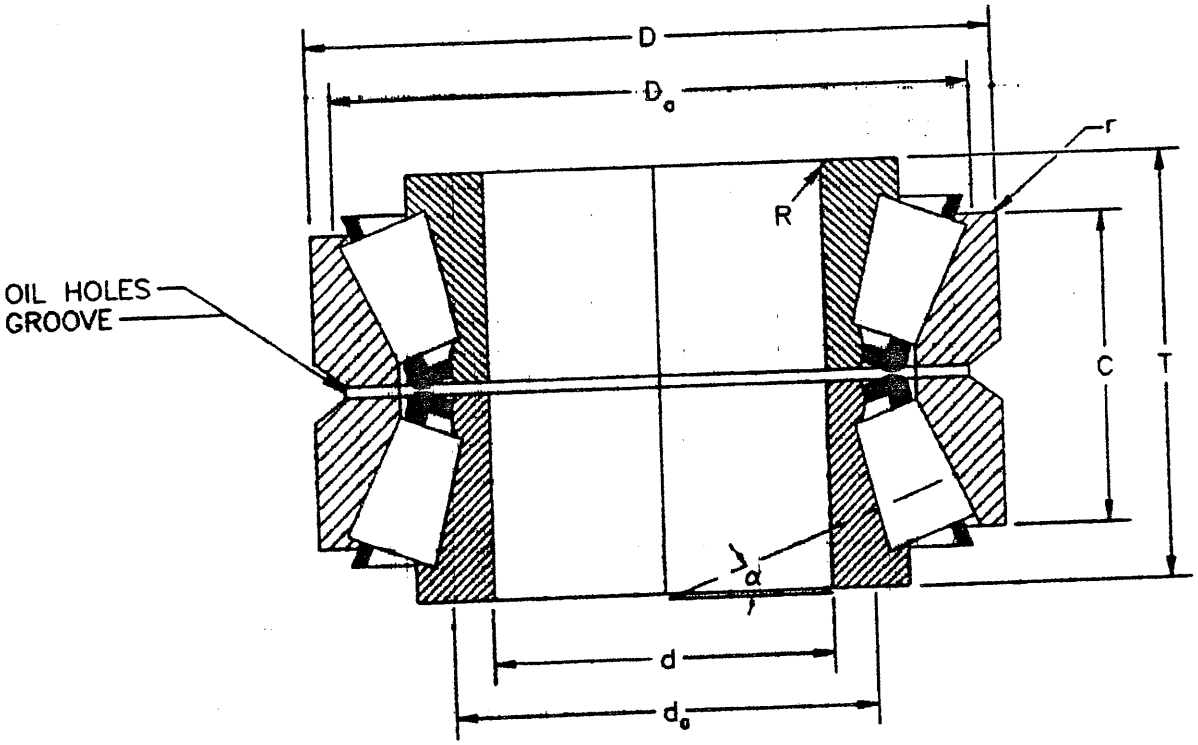
FF–B–187/9
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, DOUBLE ROW, STEEP ANGLE, TWO SINGLE CONES, ONE DOUBLE CUP, NONADJUSTABLE (TYPE TNAS)

This specification is approved by the Administrator, General Services Administration, Federal Supply Service, for the use of all federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE–CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
01 *	NA21075–21226D	0.7500	2.2500	1.4375	1.9375	0.06	0.03	1.24	2.01	.99	12800
02 *	NA43131–43319D	1.3125	3.1875	1.5625	2.1875	0.09	0.06	1.91	2.91	.87	21500
03	NA44143–44363D	1.4375	3.6250	1.5625	2.1875	0.03	0.06	2.01	3.35	.75	23300
04	NA44163–44363D	1.6250	3.6250	1.5625	2.1875	0.03	0.06	2.13	3.35	.75	13600
05 *	NA53176–53390D	1.7500	3.8750	1.7500	2.5626	0.09	0.06	2.40	3.54	.79	27700
06 *	NA55200–55444D	2.0000	4.4375	1.8125	2.5625	0.09	0.06	2.72	4.13	.66	29400
07 *	NA66212–66462D	2.1250	4.6250	2.1250	2.8750	0.14	0.03	2.87	4.37	.93	41200
08	NA72212–72488D	2.1250	4.8750	2.1875	3.0625	0.09	0.06	2.91	4.53	.79	50000
09	NA9378–9320D	3.0000	7.0000	2.9375	4.3065	0.14	0.09	4.13	6.46	.76	49100
10 *	NA98350–98789D	3.5000	7.8750	3.1581	4.5625	0.14	0.09	4.65	7.40	.92	112100
11 *	NA97450–97901D	4.5000	9.0000	3.3125	4.5625	0.14	0.09	5.51	8.38	.79	102600

* INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).

2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

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TABLE II

DASH NUMBER (REF)	BEARINGS MANUFACTURED WITH CORRECT RUNNING CLEARANCE FOR THESE FITS		
	LIVE SHAFT DIAMETER LIMITS	HOUSING	
	TIGHT FIT	LIVE SHAFT BORE LIMITS	DEAD SHAFT BORE LIMITS
		LOOSE FIT	TIGHT FIT
01	0.7515–0.7510	2.2530–2.2520	2.2490–2.2480
02	1.3140–1.3135	3.1905–3.1895	3.1865–3.1855
03	1.4390–1.4385	3.6280–3.6270	3.6240–3.6230
04	1.6265–1.6260	3.6280–3.6270	3.6240–3.6230
05	1.7515–1.7510	3.8780–3.8770	3.8740–3.8730
06	2.0020–2.0010	4.4405–4.4395	4.4365–4.4355
07	2.1270–2.1260	4.6280–4.6270	4.6240–4.6230
08	2.1270–2.1260	4.8780–4.8770	4.8740–4.8730
09	3.0025–3.0015	7.0030–7.0020	6.9990–6.9980
10	3.5030–3.5020	7.8780–7.8770	7.8740–7.8730
11	4.5035–4.5025	9.0030–9.0020	8.9990–8.9980

REQUIREMENTS:

1. Material: Cones, cups and rollers: bearing quality – carburizing grade alloy or through–hardening grade alloy steel in accordance with ASTM–A295, ASTM–A534, or ASTM–A535. Cage: carbon steel (stamped). Powdered process steel may be used.

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

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Cone bore (d)			
Size range B		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	5.0000	100	0
5.0000	12.0000	300	0

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

4. Operating temperature: Recommended operating temperature not to exceed 230°F.

5. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-I97.

6. Contact angle: All bearings are normal angle bearings having a contact angle (a) between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

7. Part number: The part number consist of the letters FFB, the basic federal specification number, the specification sheet number and a dash number from table 1.

EXAMPLE: FFB187/09 - 06

Dash Number

Federal Specification Number

8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

9. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: Inch units can be converted to soft metric units as follows:

Inch [X] 25.4 [=] millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19089-1	FFB187/09-01
MS19089-2	FFB187/09-02
MS19089-3	FFB187/09-03
MS19089-4	FFB187/09-04
MS19089-5	FFB187/09-06
MS19089-6	FFB187/09-07
MS19089-7	FFB187/09-10
MS19089-8	FFB187/09-15
MS19089-9	FFB187/09-16
MS19089-10	FFB187/09-19
MS19089-11	FFB187/09-20

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